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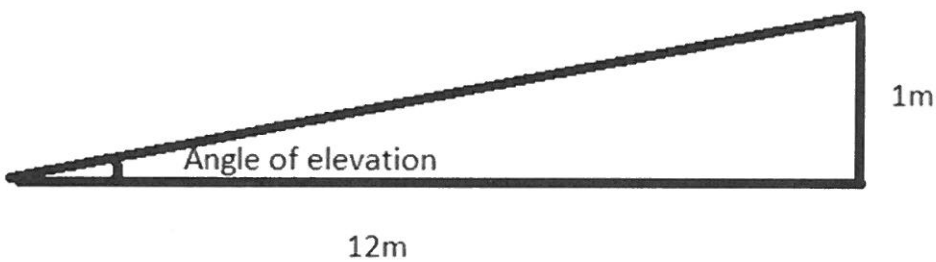
| Name:<br><br>Class: | ______________________<br><br>__________ | Date:______ |
| Baldivis<br>Secondary College | Year 10 Mathematics 2018<br>Investigation<br>Trigonometry<br>**Investigation / Validation**<br>You will need: Collaborative component:<br>Pen, calculator and a ruler | / 24<br><br>2 Lessons |

**Y10 OLNA Investigation for trigonometry.**

We will be working in groups to construct a ramp which complies with the safety standards of Western Australia.

When a slope becomes greater than **1:20 or 2.9 degrees** in inclination; we must build it according to the WA regulations, anything below is not considered a ramp. These rules specify that a maximum gradient of **1:12 or 4.8 degrees** in inclination cannot be exceeded.

**1)** Determine if the following ramp falls under the rules and if it complies: **( 3 marks )**




**Slope = rise : run = vertical height increase : Horizontal length increase**
**Gradient = rise / run = vertical height increase / Horizontal length increase**

**Show all workings**

✓ rise run

✓ 1 : 12

✓ Therefore max gradient

or ✓

$\tan\theta = \frac{opp}{adj}$

$\theta = \tan^{-1}\left(\frac{opp}{adj}\right)$ ✓

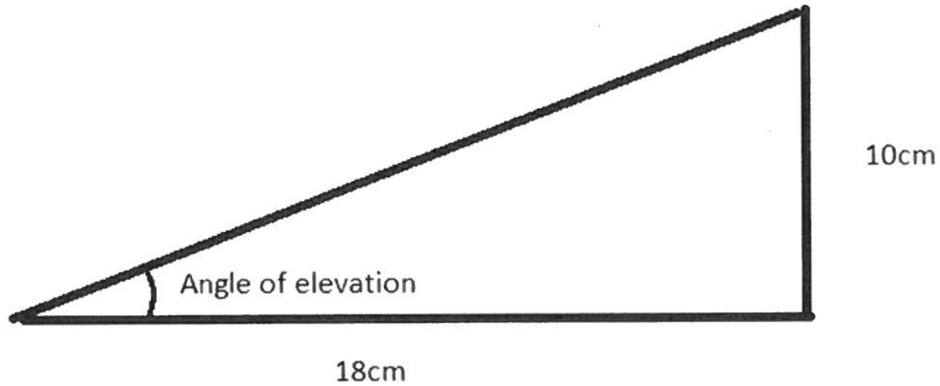
$= \tan^{-1}\left(\frac{1}{12}\right)$ ✓

$= 4.76^\circ$ ✓

Therefore less than max gradient

( **3 marks** )

**2)** Tom decides to place plywood down the stairs to make a wheelchair access. Is the following rap considered safe? Explain why or why not? Show all workings.




**Show all workings**

$\theta = \tan^{-1}\left(\frac{opp}{adj}\right)$ ✓

$= \tan^{-1}\left(\frac{10}{18}\right)$

$= 29.1^\circ$ ✓

✓

$29.1^\circ > 4.8^\circ$ ∴ Steeper

0 ✓

$10 \div 18 = 0.55$ ✓

$1 \div 12 = 0.0833$

✓

$0.55 > 0.0833$

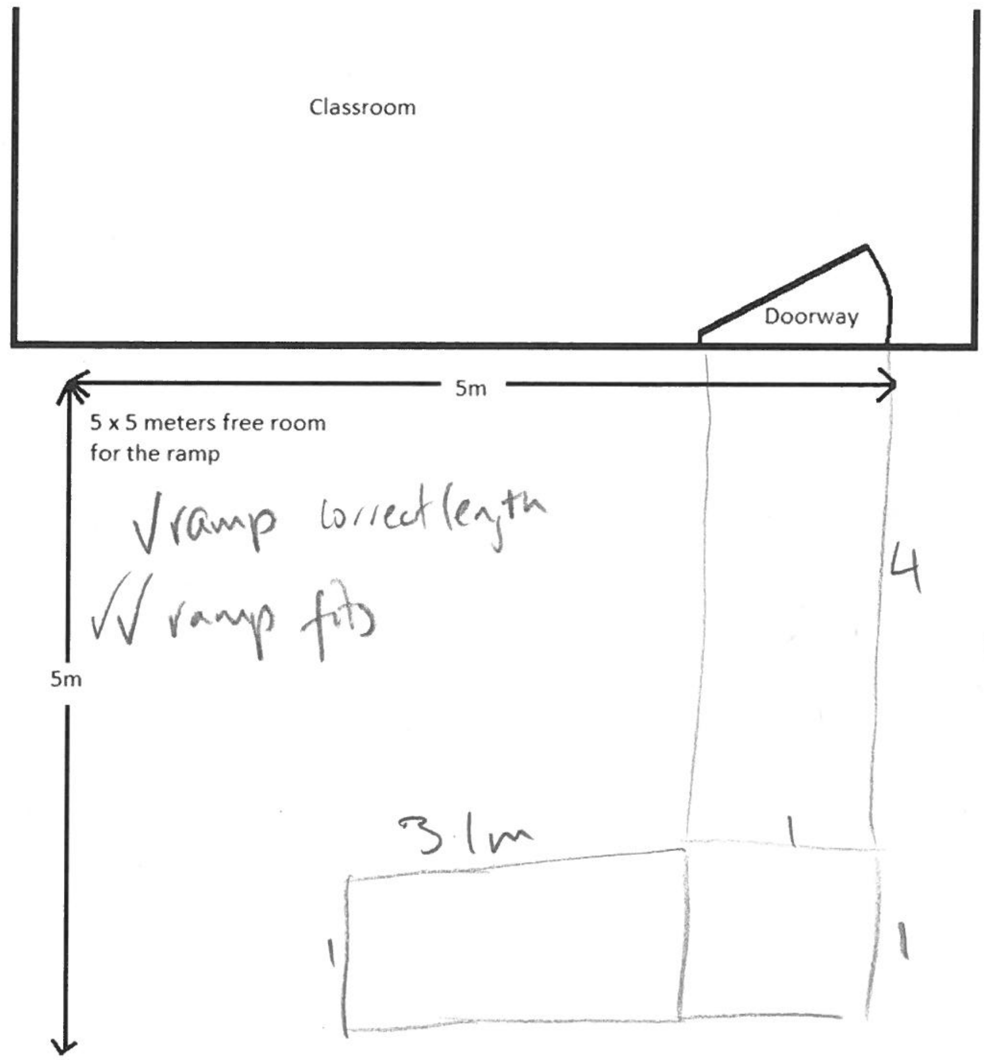
∴ Steeper ✓

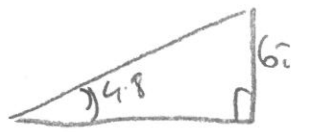
**( 6 marks )**

**3)** A classroom is 60 cm above the surrounding ground level. Please construct a ramp to allow wheelchair accessibility given there is only 5×5 meters of room in front to make the ramp. Please give birds eye view and side view with all measurements: If you do not have enough room to make the ramp, make half the ramp, then leave a metre flat landing around the bend and go back to double the length. Show all workings.

**Birdseye view:**

**Side** **view:**




$\checkmark$ $adj = \frac{opp}{\tan\theta}$

$= \frac{60}{\tan 4.8}$

$= 714cm$

$\checkmark\checkmark = 7.1m$

**4)** Students are to go outside, **observe** and **measure** the concrete ramp up to the D block, as well go and look at the 2 science demount-able rooms which have a wooden ramp that makes a bend.

Students are to measure the slope of both.

**Slope of D block ramp:** **( 2 marks )**

$\theta = \tan^{-1}\left(\frac{opp}{adj}\right) = \tan^{-1}\left(\frac{30}{565}\right) = 3.0^\circ$ $\checkmark$ $\checkmark$

**Slope of Science-D:** **( 2 marks )**

$\theta = \tan^{-1}\left(\frac{45}{600}\right) = 4.3^\circ$

✓Formula ✓Answer $< 4.8^\circ$

**( 8 marks )**

**5)** Constructing our ramp:
There are two options for making our ramps.

**<u>Option 1</u>** is to pour a concrete version of our ramp, which involves making a wooden cheap frame, adding iron bars and footings and pour concrete.

**<u>Option 2</u>** is to make footings and attach a wooden frame and laying planks which are attached to the frame.

**(I) For <u>option 1</u>**, please calculate how much concrete is needed if the ramp is 1m wide and 0.15m thick.

<u>Volume = Length of ramp (m) x Ramp thickness (m) x ramp width (m)</u>

$7.14 \times 1 \times 0.15 = 1.05\,m^3$

✓ ✓ ✓ ✓

**(II) For <u>option 2</u>,** Please calculate how many wooden beams are needed to cover the frame if they are 1m long, 14cm wide, and have a few millimetre gap between beams. Therefore each metre of pathway uses 7 beams.

<u>Number of beams = length of ramp (m) x 7</u>

$7.14 \times 7 = 49.98$ ✓

✓ ✓ $= 50$ ✓ beams